Python Assessment

Task 1

Given an integer, , perform the following conditional actions:

- If is odd, print Weird
- If is even and in the inclusive range of to , print Not Weird
- If is even and in the inclusive range of to , print Weird
- If is even and greater than , print Not Weird

Input Format-

A single line containing a positive integer, .

Constraints

- 1<= n<=100

Output Format

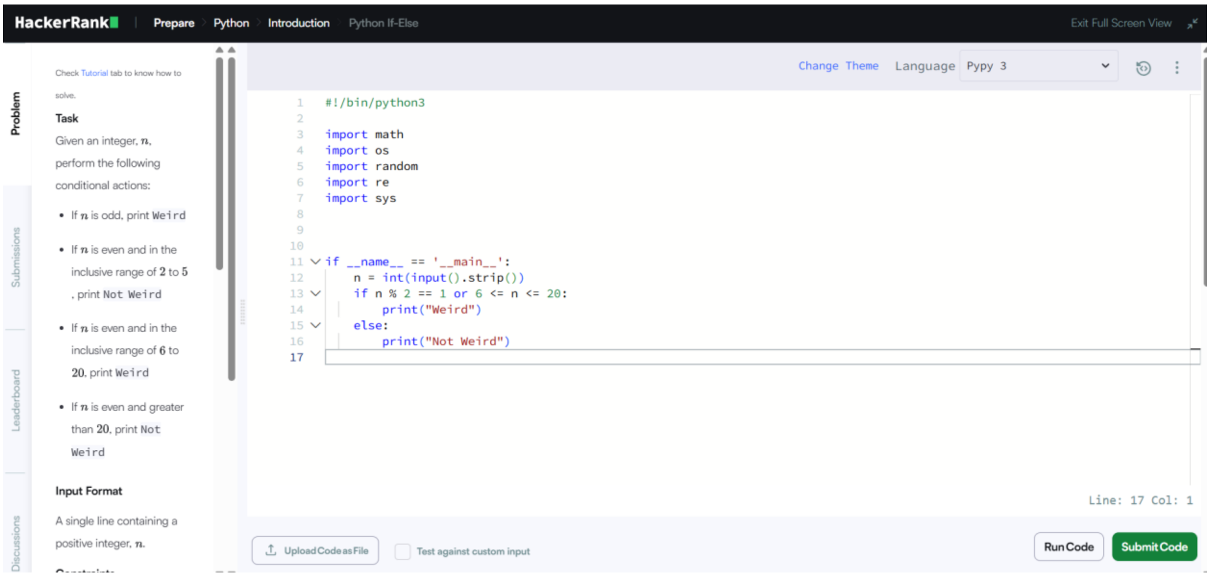
Print Weird if the number is weird. Otherwise, print Not Weird.

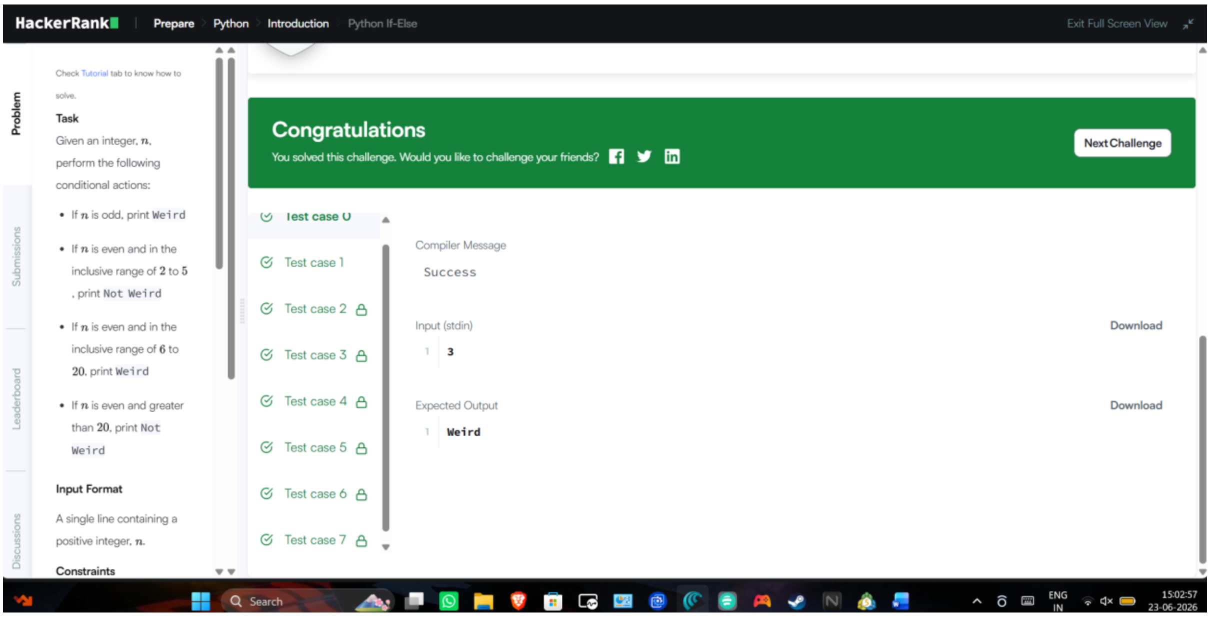
Sample Input 0

3

Sample Output 0

Weird





Task 2

The provided code stub reads two integers from STDIN, and . Add code to print three lines where:

1. The first line contains the sum of the two numbers.
2. The second line contains the difference of the two numbers (first - second).
3. The third line contains the product of the two numbers.

Example

Print the following:

8

-2

15

Input Format

The first line contains the first integer, .
The second line contains the second integer, .

Constraints

$1 \leq a \leq 10^{10}$
$1 \leq b \leq 10^{10}$

Output Format

Print the three lines as explained above.

Sample Input 0

3

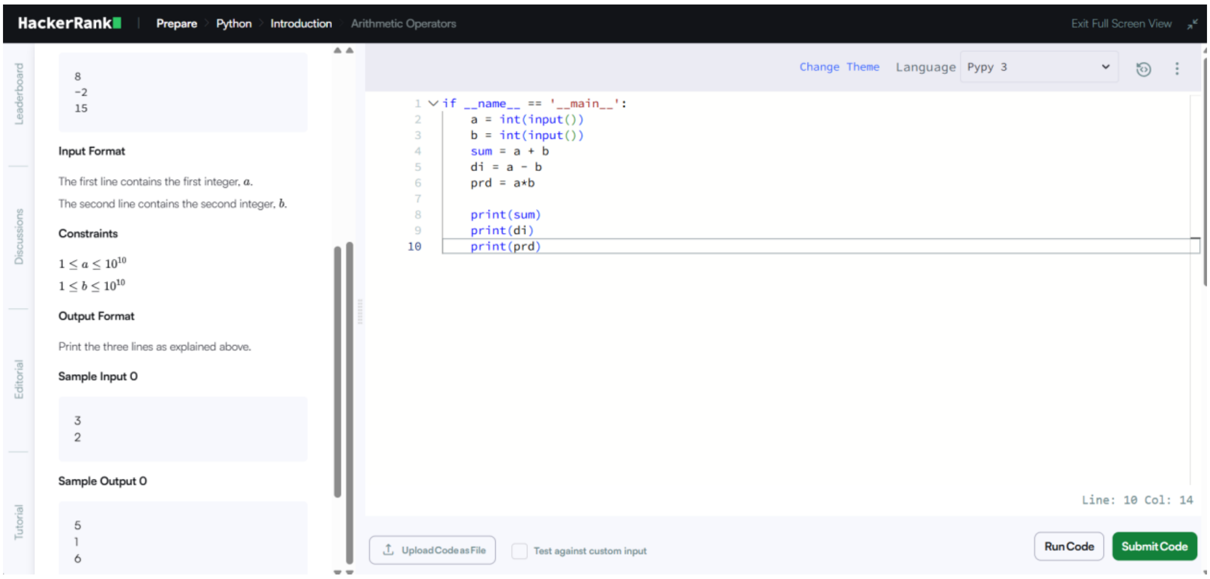
2

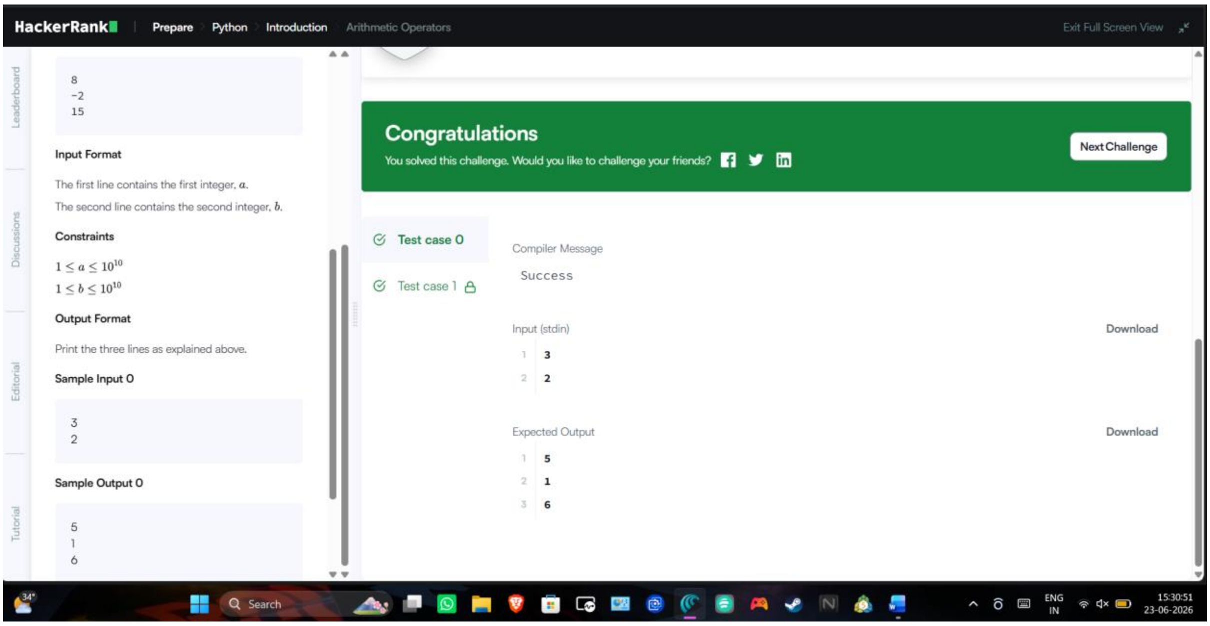
Sample Output 0

5

1

6





Python Assignment

Section 1

1. Print your name, age, and city in a single formatted sentence

```
name = str(input("enter your name:"))
age = int(input("enter your age:"))
city = str(input("enter your city"))
print(f"My name is {name}, I am {age} years old, and I live in {city}.")
```

```
enter your name:Vijay
enter your age:21
enter your citychennai
My name is Vijay, I am 21 years old, and I live in chennai.
```

2. Print numbers from 1 to 10 in a single line separated by space.

```
n = int(input("enter the range number"))
for i in range(1,n+1):
  print(i,end=" ")
```

```
enter the range number10
1 2 3 4 5 6 7 8 9 10
```

3. Print a multiplication table for a given number (e.g., 7).

```
n = int(input("enter a number to print multiplication table:"))
m = 10
for i in range (1,m+1):
  print(n,"x",i,"=",n*i)
```

```
enter a number to print multiplication table:7
7 x 1 = 7
7 x 2 = 14
7 x 3 = 21
7 x 4 = 28
7 x 5 = 35
7 x 6 = 42
7 x 7 = 49
7 x 8 = 56
7 x 9 = 63
7 x 10 = 70
```

4. Print a right-angle triangle star pattern (5 rows).

```
row = int(input("enter the number of rows:"))
for i in range(1,row+1):
  print("^" * i)
```

```
enter the number of rows:5
^
^^
^^^
^^^^
^^^^^
```

5. Print the sum of first 20 natural numbers.

```
n = int(input("enter a natural number to find their sum:"))
sum = 0
for i in range(1,n+1):
  sum = sum+i

print(sum)
```

```
enter a natural number to find their sum:20
210
```

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6. Print all odd numbers between 1 and 50.

```
n = 1
m = 50

for i in range(1,50):
  if(i%2!=0):
    print(i,end=" ")
```

```
1 3 5 7 9 11 13 15 17 19 21 23 25 27 29 31 33 35 37 39 41 43 45 47 49
```

7. Print a string in reverse using print statement logic

```
name = str(input("enter your name:"))
print(name[::-1])
```

```
enter your name:vijay
yajiv
```

8. Print two strings on the same line using a custom separator.

```
n = str(input("enter a string:"))
m = str(input("enter next string:"))
print(n,"&",m)
```

```
enter a string:vijay
```


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7. Print a string in reverse using print statement logic

```
name = str(input("enter your name:"))
print(name[::-1])
```

```
enter your name:vijay
yajiv
```

8. Print two strings on the same line using a custom separator.

```
n = str(input("enter a string:"))
m = str(input("enter next string:"))
print(n,"&",m)
```


```
enter a string:vijay
enter next string:sanjay
vijay & sanjay
```

9. Print output using escape characters (newline, tab)

```
n = str(input("enter a string:"))
m = str(input("enter the second string:"))
print(f"{n} \n {m}")
print(f"{n} \t {m}")
```

```
enter a string:vijay
```


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9. Print output using escape characters (newline, tab)

```
n = str(input("enter a string:"))
m = str(input("enter the second string:"))
print(f"{n} \n {m}")
print(f"{n} \t {m}")
```


```
enter a string:vijay
enter the second string:sanjay
vijay
 sanjay
vijay    sanjay
```

10. Print the result of arithmetic operations inside a formatted string.

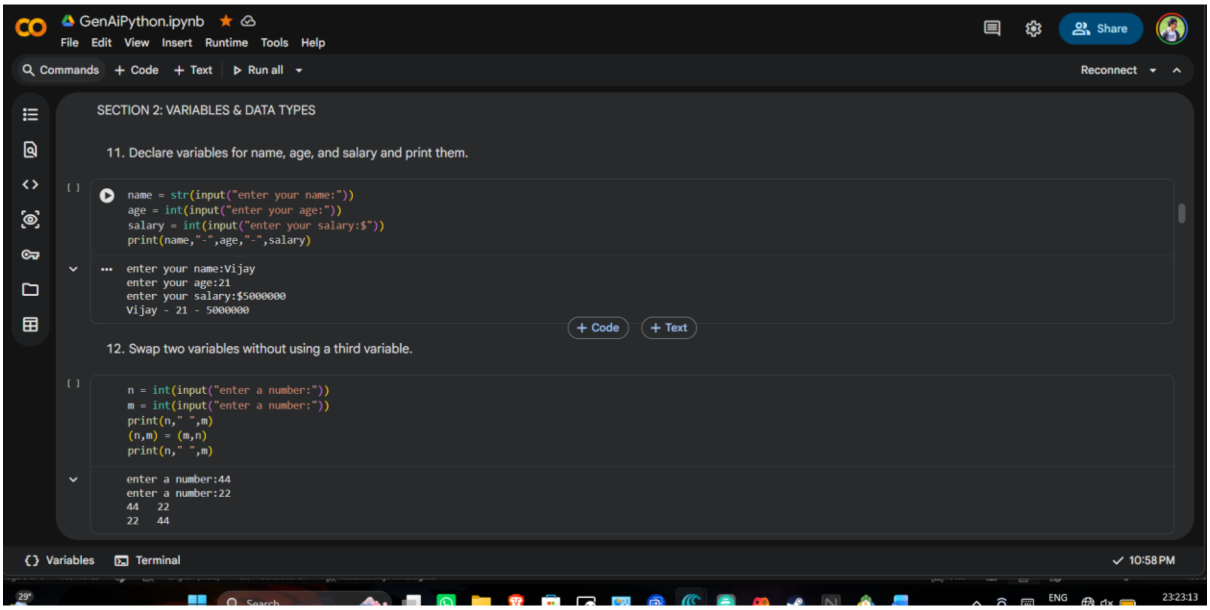
```
a = int(input("enter a number:"))
b = int(input("enter a number:"))
print(f"Addition ={a+b},sub={a-b},Multi={a*b},div={a/b}")
```

```
enter a number:5
enter a number:1
Addition =6,sub=4,Multi=5,div=5.0
```

SECTION 2: VARIABLES & DATA TYPES

11. Declare variables for name, age, and salary and print them.

Section 2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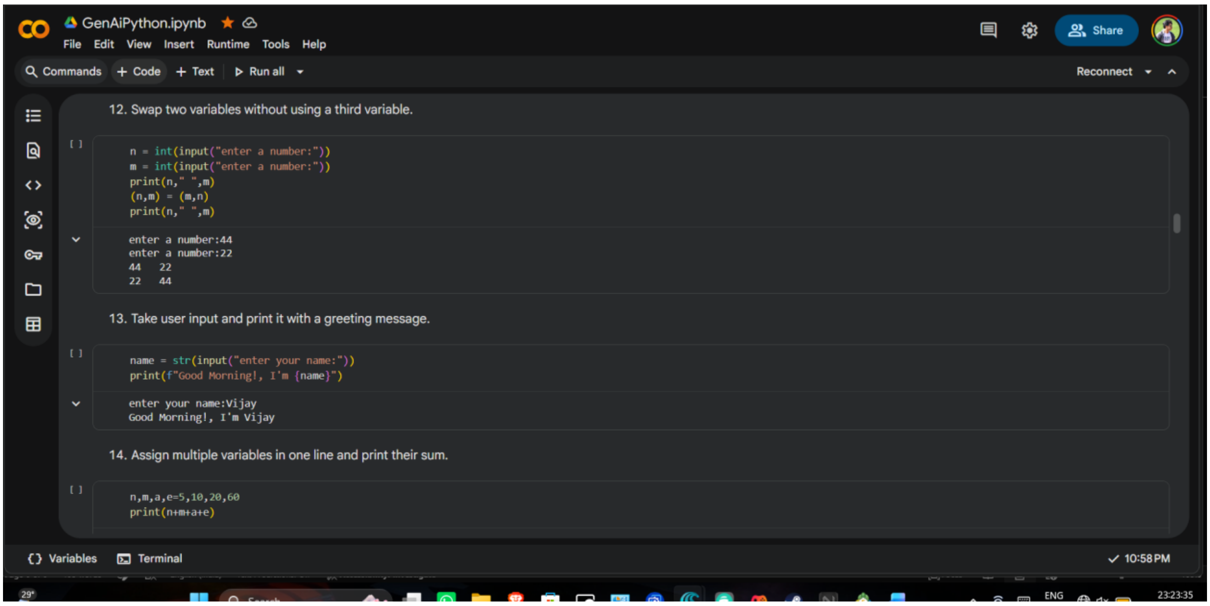
SECTION 2: VARIABLES & DATA TYPES
11. Declare variables for name, age, and salary and print them.
name = str(input("enter your name:"))
age = int(input("enter your age:"))
salary = int(input("enter your salary:$"))
print(name,"-",age,"-",salary)
enter your name:Vijay
enter your age:21
enter your salary:$5000000
Vijay - 21 - 5000000
+ Code + Text
12. Swap two variables without using a third variable.
n = int(input("enter a number:"))
m = int(input("enter a number:"))
print(n," ",m)
(n,m) = (m,n)
print(n," ",m)
enter a number:44
enter a number:22
44 22
22 44
Variables Terminal
10:58 PM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Reconnect
12. Swap two variables without using a third variable.
n = int(input("enter a number:"))
m = int(input("enter a number:"))
print(n," ",m)
(n,m) = (m,n)
print(n," ",m)
enter a number:44
enter a number:22
44 22
22 44
13. Take user input and print it with a greeting message.
name = str(input("enter your name:"))
print(f"Good Morning!, I'm {name}")
enter your name:Vijay
Good Morning!, I'm Vijay
14. Assign multiple variables in one line and print their sum.
n,m,a,e=5,10,20,60
print(n+m+a+e)
Variables Terminal
10:58 PM

14. Assign multiple variables in one line and print their sum.

```
n,m,a,e=5,10,20,60
print(n+m+a+e)
```

```
95
```

15. Check and print the data type of a variable.

```
a = int(input("enter a number:"))
b = str(input("enter a word:"))
c = float(input("enter a decimal number:"))
d = bool(input("enter true or false:"))
print(type(c))
print(type(d))
print(type(a))
print(type(b))
```

```
enter a number:10
enter a word:fine
enter a decimal number:10.7
enter true or false:true
<class 'float'>
<class 'bool'>
<class 'int'>
<class 'str'>
```

16. Convert a string number into an integer and perform addition

```
a = str(input("enter a number:"))
a = int(a)
print(a+a)
```

```
enter a number:5
10
```

17. Convert an integer into a string and concatenate with another string.

```
a = int(input("enter a number:"))
b = str(input("enter a string:"))
print(b+str(a))
```

```
enter a number:400
enter a string:vijay
vijay400
```

18. Create variables of int, float, string, and boolean and print their types

```
a = int(input("enter a number:"))
b = float(input("enter a decimal number:"))
c = str(input("enter a word:"))
d = bool(input("enter true or false:"))
print(type(a))
```

18. Create variables of int, float, string, and boolean and print their types

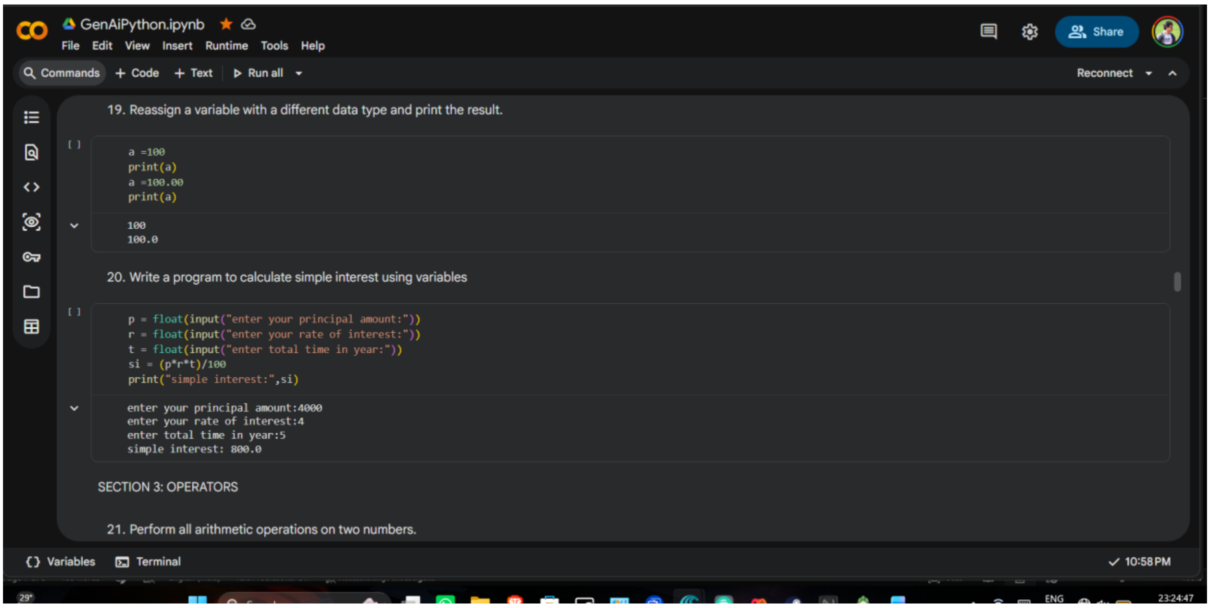
```
a = int(input("enter a number:"))
b = float(input("enter a decimal number:"))
c = str(input("enter a word:"))
d = bool(input("enter true or false:"))
print(type(a))
print(type(b))
print(type(c))
print(type(d))
```

```
enter a number:500
enter a decimal number:89.00
enter a word:faith
enter true or false:false
<class 'int'>
<class 'float'>
<class 'str'>
<class 'bool'>
```

19. Reassign a variable with a different data type and print the result.

```
a =100
print(a)
a =100.00
print(a)
```

```
100
```



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19. Reassign a variable with a different data type and print the result.

```
a =100
print(a)
a =100.00
print(a)
```

```
100
100.0
```

20. Write a program to calculate simple interest using variables

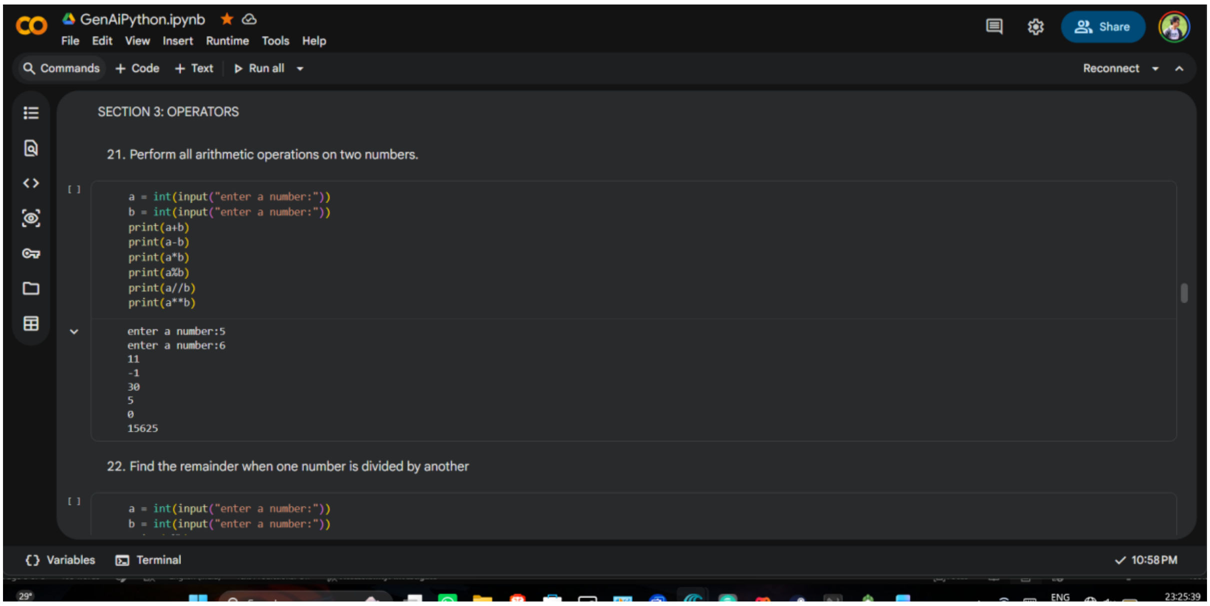
```
p = float(input("enter your principal amount:"))
r = float(input("enter your rate of interest:"))
t = float(input("enter total time in year:"))
si = (p*r*t)/100
print("simple interest:",si)
```

```
enter your principal amount:4000
enter your rate of interest:4
enter total time in year:5
simple interest: 800.0
```

SECTION 3: OPERATORS

21. Perform all arithmetic operations on two numbers.

Section 3



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SECTION 3: OPERATORS

21. Perform all arithmetic operations on two numbers.

```
a = int(input("enter a number:"))
b = int(input("enter a number:"))
print(a+b)
print(a-b)
print(a*b)
print(a%b)
print(a//b)
print(a**b)
```

```
enter a number:5
enter a number:6
11
-1
30
5
0
15625
```

22. Find the remainder when one number is divided by another

```
a = int(input("enter a number:"))
b = int(input("enter a number:"))
```

22. Find the remainder when one number is divided by another

```
a = int(input("enter a number:"))
b = int(input("enter a number:"))
print(a%b)
```

```
enter a number:45678
enter a number:32
14
```

23. Calculate the power of a number using an operator

```
a = int(input("enter the number:"))
print(a**2)
```

```
enter the number:5
25
```

24. Use floor division to divide two numbers and print the result.

```
a = int(input("enter a number:"))
b = int(input("enter a number:"))
print(a//b)
```

24. Use floor division to divide two numbers and print the result.


```
a = int(input("enter a number:"))
b = int(input("enter a number:"))
print(a//b)
```


```
enter a number:10
enter a number:2
5
```

25. Compare two numbers and print which one is greater

```
a = int(input("enter a number:"))
b = int(input("enter a number:"))
if(a > b):
  print("a is greater than b",a)
else:
  print("b is greater than a",b)
```

```
enter a number:100
enter a number:500
b is greater than a 500
```

26. Check if a number is even or odd using operators.

26. Check if a number is even or odd using operators.

```
a = int(input("enter a number:"))
if( a%2==0):
  print("even")
else:
  print("odd")
```

```
enter a number:44
even
```

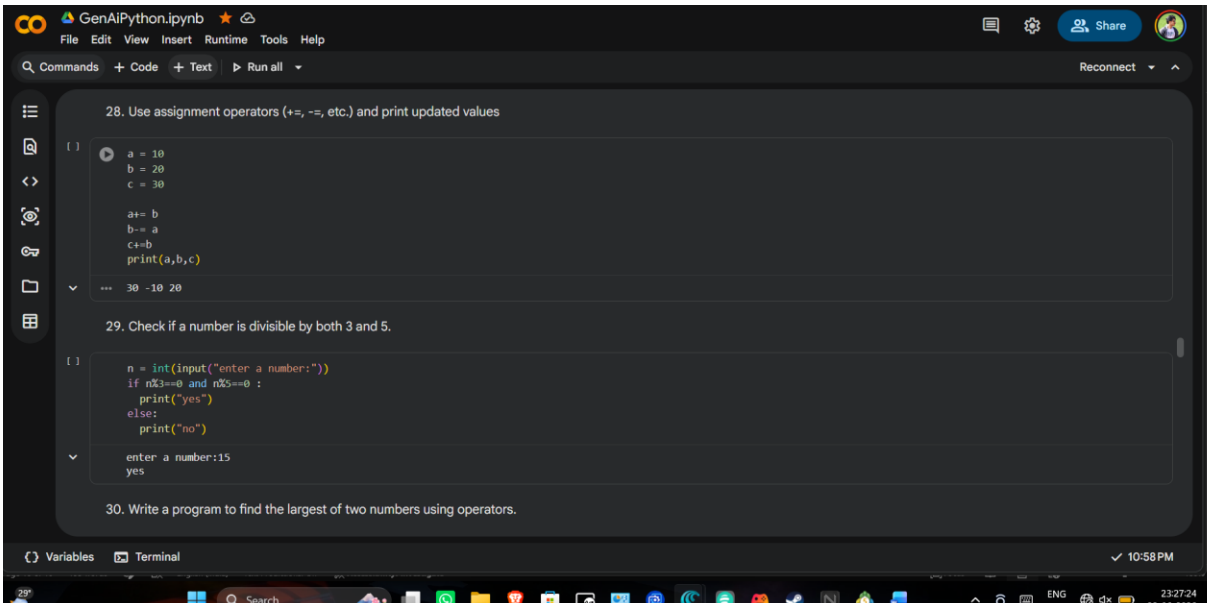
27. Use logical operators to check if a number lies between 10 and 50.

```
n = int(input("enter a number:"))


if n >=10 and n <=50:
  print("the number lieas in between")
else:
  print("the number does't")
```

```
enter a number:44
the number lieas in between
```

28. Use assignment operators (+=, -=, etc.) and print updated values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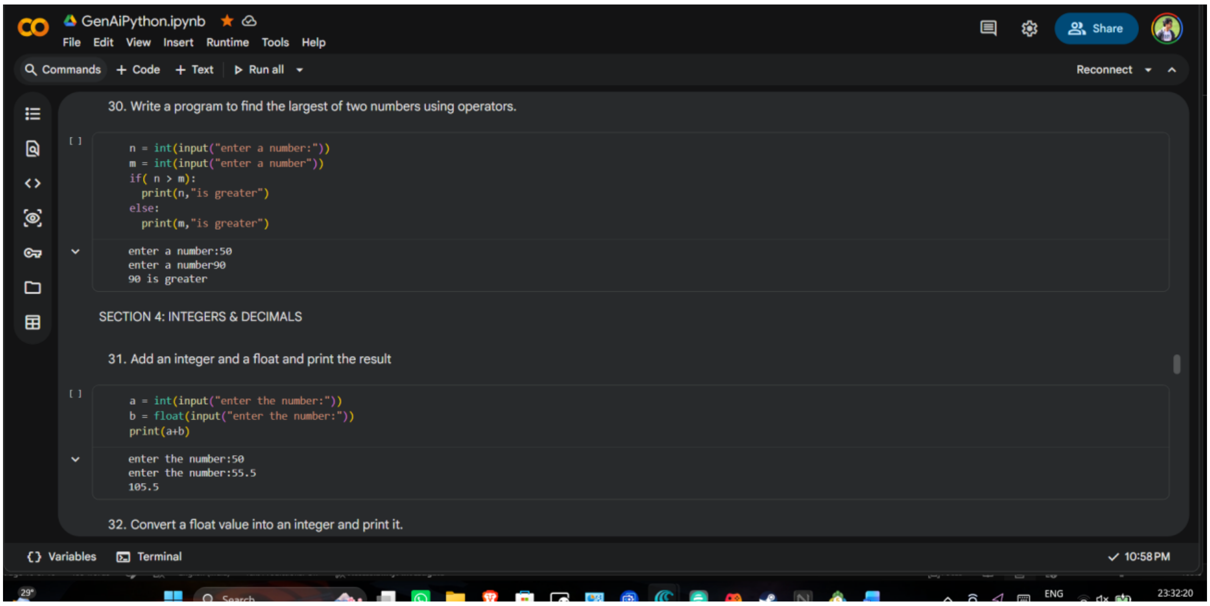
28. Use assignment operators (+=, -=, etc.) and print updated values
a = 10
b = 20
c = 30
a+= b
b-= a
c+=b
print(a,b,c)
30 -10 20
29. Check if a number is divisible by both 3 and 5.
n = int(input("enter a number:"))
if n%3==0 and n%5==0 :
print("yes")
else:
print("no")
enter a number:15
yes
30. Write a program to find the largest of two numbers using operators.
Variables
Terminal
10:58 PM

Section 4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Reconnect
30. Write a program to find the largest of two numbers using operators.
n = int(input("enter a number:"))
m = int(input("enter a number"))
if(n > m):
print(n,"is greater")
else:
print(m,"is greater")
enter a number:50
enter a number90
90 is greater
SECTION 4: INTEGERS & DECIMALS
31. Add an integer and a float and print the result
a = int(input("enter the number:"))
b = float(input("enter the number:"))
print(a+b)
enter the number:50
enter the number:55.5
105.5
32. Convert a float value into an integer and print it.
Variables
Terminal
10:58 PM

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32. Convert a float value into an integer and print it.

```
a = 55.5
a = int(a)
print(a)
```

```
55
```

33. Divide two numbers and print the decimal result.

```
a = int(input("type a number:"))
b = int(input("give a number:"))
print(float(a/b))
```

```
type a number:325
give a number:62
5.241935483870968
```

34. Round a decimal number to 2 decimal places.

```
n = float(input("enter the number:"))
print(round(n,2))
```

```
enter the number:55.5555
55.56
```

Variables Terminal 10:58 PM

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35. Format a float to display only 3 decimal points.

```
a = float(input("enter a number:"))
print(round(a,3))
```

```
enter a number:3.337859
3.338
```

36. Check whether a variable is of type integer.

```
n = 59
if type(n) == int:
  print("it is integer")
else:
  print("it is not a integer")
```

```
it is integer
```

37. Find the absolute value of a negative number

```
a = int(input("enter a negative number:"))
print("Absolute number",abs(a))
```

```
enter a negative number:-99
Absolute number 99
```

Variables Terminal 10:58 PM

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38. Multiply two decimal numbers and print the result.

```
n = float(input("enter a decimal number:"))
m = float(input("enter a decimal number:"))
print(n*m)
```

```
enter a decimal number:57.3
enter a decimal number:34.2
1959.66
```

39. Convert a decimal number into a string.

```
n =55.7
print(type(n))
n = str(n)
print(type(n))
```

```
<class 'float'>
<class 'str'>
```

40. Calculate the average of three numbers (including decimals).

```
a = int(input("enter the 1st number:"))
b = int(input("enter the 2nd number:"))
c = int(input("enter the 3rd nummber:"))
avg = float((a + b + c)/3)
```

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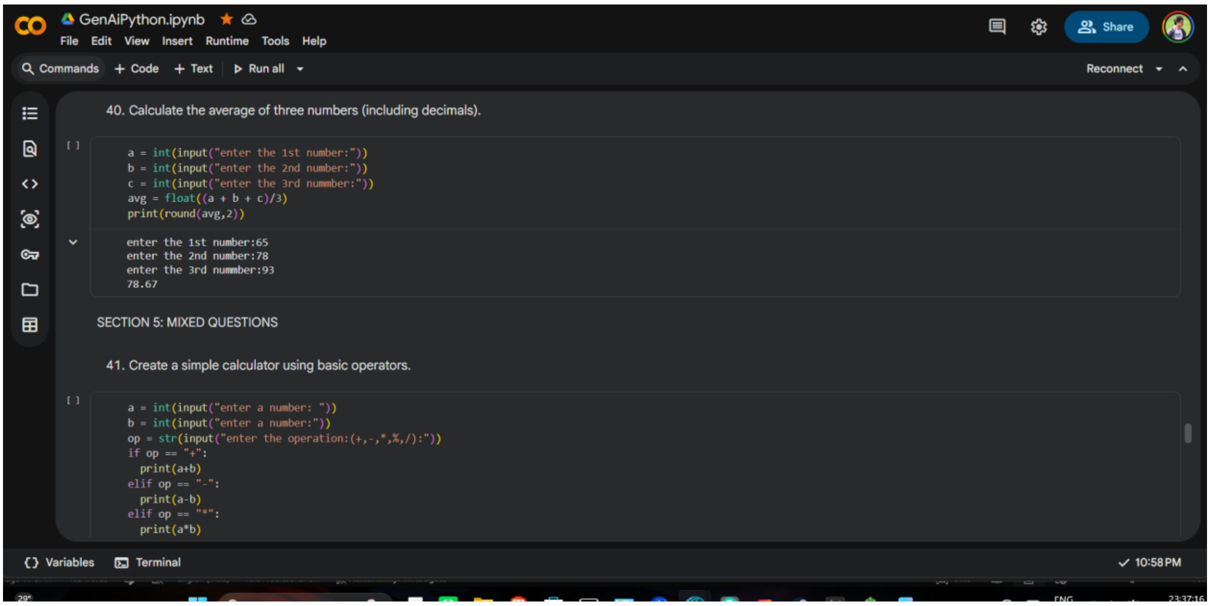

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40. Calculate the average of three numbers (including decimals).

```
a = int(input("enter the 1st number:"))
b = int(input("enter the 2nd number:"))
c = int(input("enter the 3rd nummber:"))
avg = float((a + b + c)/3)
print(round(avg,2))
```

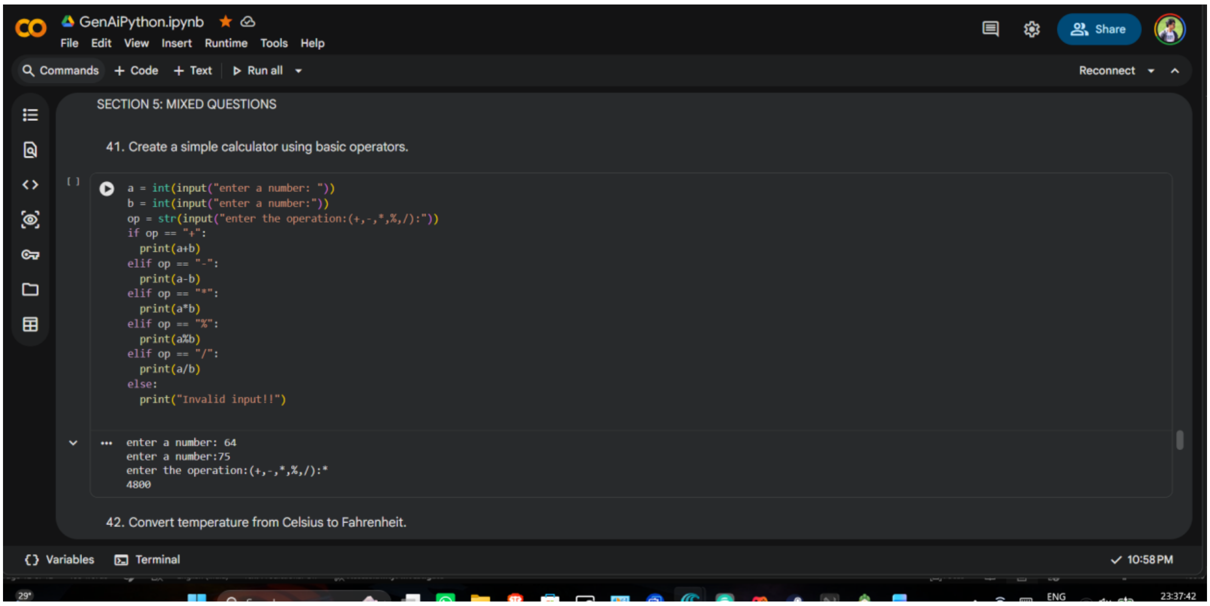
```
enter the 1st number:65
enter the 2nd number:78
enter the 3rd nummber:93
78.67
```

SECTION 5: MIXED QUESTIONS

41. Create a simple calculator using basic operators.

```
a = int(input("enter a number: "))
b = int(input("enter a number:"))
op = str(input("enter the operation:(+,-,*,%,/):"))
if op == "+":
  print(a+b)
elif op == "-":
  print(a-b)
elif op == "*":
  print(a*b)
```

Section 5



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SECTION 5: MIXED QUESTIONS

41. Create a simple calculator using basic operators.

```
a = int(input("enter a number: "))
b = int(input("enter a number:"))
op = str(input("enter the operation:(+,-,*,%,/):"))
if op == "+":
  print(a+b)
elif op == "-":
  print(a-b)
elif op == "*":
  print(a*b)
elif op == "%":
  print(a%b)
elif op == "/":
  print(a/b)
else:
  print("Invalid input!!")
```

```
enter a number: 64
enter a number:75
enter the operation:(+,-,*,%,/):*
4800
```

42. Convert temperature from Celsius to Fahrenheit.

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42. Convert temperature from Celsius to Fahrenheit.

```
c = int(input("enter the celsius:"))
f = (c * 1.8)+32
print(int(f))
```

```
enter the celsius:100
212
```

43. Check whether a number is positive, negative, or zero

```
n = int(input("enter a number:"))
if n == 0:
  print("it's zero")
elif n > 0:
  print("it's positive")
else:
  print("it's negative")
```

```
enter a number:-4
it's negative
```

44. Find the sum of digits of a number

```
n = int(input("enter a number to find it's sum:"))
```

Variables Terminal 10:58 PM

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44. Find the sum of digits of a number

```
n = int(input("enter a number to find it's sum:"))
sum =0
for i in range(1,n+1):
  sum = sum + i

print(f"the sum of{n} is {sum}")
```

```
enter a number to find it's sum:3
the sum of3 is 6
```

45. Calculate the area of a circle using variables and float values.

```
r = int(input("enter the radius of circle:"))
area = float(3.14 * r * r)
print(f"The area of the circle is {area}")
```

```
enter the radius of circle:5
The area of the circle is 78.5
```

46. Find the square and cube of a number

```
n = int(input("enter a number to find it's square and cube:"))
square = n*n
cube = n**3
```

Variables Terminal 10:58 PM

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Commands + Code + Text Run all Reconnect

46. Find the square and cube of a number

```
n = int(input("enter a number to find it's square and cube:"))
square = n*n
cube = n**3
print(f"square is {square} and the cube is {cube}")
```

```
enter a number to find it's square and cube:2
square is 4 and the cube is 8
```

47. Write a program to calculate the perimeter of a rectangle.

```
l = int(input("enter the length of the rectangle:"))
b = int(input("enter the breadth of the rectangle:"))
peri = 2*(l+b)
print(peri)
```

```
enter the length of the rectangle:5
enter the breadth of the rectangle:4
18
```

48. Take two decimal inputs and print their product.

```
a = float(input("enter the decimal number:"))
b = float(input("enter the decimal number:"))
print(a*b)
```

Variables Terminal 10:58 PM

48. Take two decimal inputs and print their product.

```python
a = float(input("enter the decimal number:"))
b = float(input("enter the decimal number:"))
print(a*b)
```

```
enter the decimal number:5.6
enter the decimal number:2.2
12.32
```

49. Check if two numbers are equal using comparison operators.

```python
n = int(input("number1:"))
m = int(input("number2:"))
if n == m:
  print("equal")
else:
  print("not equal")
```

```
number1:50
number2:45
not equal
```

50. Write a program to calculate the average of 5 user-input numbers

50. Write a program to calculate the average of 5 user-input numbers

```python
n = int(input("enter the number of user input:"))
avg = 0
for i in range(1,n+1):
  m = int(input("enter the number the number:"))
  avg = avg + m
avg = avg / n
print(avg)
```

```
enter the number of user input:5
enter the number the number:3
enter the number the number:4
enter the number the number:7
enter the number the number:8
enter the number the number:9
6.2
```